Senith Jayakody

Electrical and Electronic Engineering Graduate – University of Peradeniya

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Research Interests

My research interests include **signal processing, statistical learning, and machine learning**; **image processing, computational imaging, and generative modeling**; and **intelligent sensing systems**.

Education

B.Sc. Eng. (Hons.) in Electrical and Electronic Engineering *2021 – 2025*

University of Peradeniya, Sri Lanka GPA: **3.85/4.00**

First Class Honours | **Top 6% of Graduating Class (115 Students)** | Academic Transcript

G.C.E. Advanced Level (Sri Lankan University Entrance Examination) *2019*

Combined Mathematics (A), Physics (A), Chemistry (A)

Top 3% Nationwide (Physical Science Stream) | Result Sheet

Publications

Journal Articles and Manuscripts

[J1] Tilakarathna, U.[†], **Jayakody, S.**[†], Jayasooriya, K., Godaliyadda, R., Ekanayake, M. P., Nawinne, I., and Rathnayake, C., "Empirical mode decomposition and interpretable machine learning for preterm birth classification from electrohysterography," Manuscript under review at *PLOS ONE*, 2026. arXiv: 2608.17643

[J2] Ranasinghe, G. A. S. L.[†], **Jayakody, J. A. S. T.**[†], De Silva, M. C. L., Thilakarathne, G., Godaliyadda, G. M. R. I., Herath, H. M. V. R., Ekanayake, M. P. B., and Navaratnarajah, S. K., "Reflectance multispectral imaging for soil grain-size composition estimation and USDA texture classification," *IEEE Access*, 2026. DOI: 10.1109/ACCESS.2026.3721276

[†] Equal contribution.

Conference Papers and Submissions

[C1] Perera, S., Ranasinghe, S.**, Jayakody, S.**, Epakanda, B., Godaliyadda, R., and Ekanayake, M. P., "Feature conditioned diffusion for audio generation," in *2025 IEEE 19th International Conference on Industrial and Information Systems (ICIIS)*, 2026, pp. 162–167. DOI: 10.1109/ICIIS69028.2026.11450741

[C2] Ranasinghe, S.**, Jayakody, S.**, Silva, M. D., Herath, V., Godaliyadda, R., Ekanayake, M. P., Navaratnarajah, S., Kizel, F., and Thilakarathne, G., "Deep learning for soil moisture content estimation via reflectance multispectral imaging," in *2024 International Conference on Advances in Technology and Computing (ICATC)*, 2024, pp. 1–6. DOI: 10.1109/ICATC64549.2024.11025290

[C3] Apakanda, B., Ranasinghe, S.**, Jayakody, S.**, Appuhami, H. D. M. D., Godaliyadda, R., Ekanayake, M. P., and Ragel, R., "Colour-INR: Implicit neural representation for image colorization through colour guidance," Under review at the 15th International Conference on Learning Representations (ICLR 2027), 2026

Research and Professional Experience

AI Research and Development Engineer *Aug 2026 – Present*

FARBE Technologies (Pvt) Ltd Sri Lanka

- Developing and evaluating *AI-based signal and image processing methods* for research and industry.

Research Assistant *Aug 2025 – Jul 2026*

Multidisciplinary AI Research Centre (MARC), University of Peradeniya Peradeniya, Sri Lanka

- Conducted research in *biomedical signal processing, machine learning, generative modeling, and computational imaging*, contributing to experimental design, model evaluation, and research publications.
- Guided junior researchers in experimental workflows, data analysis, and manuscript preparation.

Research and Development Intern *Jul 2024 – Oct 2024*

Swartup Pvt. Ltd. Sri Lanka

- Contributed to embedded-system development using *nRF5340, Zephyr RTOS, LoRa, and GSM technologies*.

Research and Development Intern *Jul 2023 – Oct 2023*

SLT Digital Lab, Sri Lanka Telecom PLC Sri Lanka

- Developed *IoT-based environmental monitoring systems* integrating ESP32, MQTT, Firebase, and sensors.

Selected Research Projects

01. Preterm Birth Prediction from EHG and Tocographic Signals *2025 – Present*

- Developed *EMD-based signal decomposition and interpretable feature-extraction pipelines* for EHG signals, evaluating intrinsic mode functions, burst morphology, temporal energy, and entropy features.
- Demonstrated that *annotation-independent, fixed 3-minute segmentation* achieved higher mean recording-level classification performance than annotated-interval segmentation in exploratory experiments, suggesting useful information beyond annotated contractions.
- Conducted *recording-level effect-size analysis and grouped permutation feature importance*, identifying distinct preterm-associated signal patterns and temporal-energy features as the strongest feature family in the selected models.

Research Outputs: EMD preprint Code

02. Colour-INR: Image Colorization using Implicit Neural Representations *2026 – Present*

- Co-developed *Colour-INR*, an image-specific implicit neural representation using SIREN to reconstruct chrominance from only **0.5% sparse colour hints**, without pretrained colourization models.
- Investigated *edge-aware chrominance regularization* to preserve image boundaries, improving mean PSNR from **29.80 to 32.64 dB (+2.84 dB)** over sparse-hint supervision alone on Kodak24.
- Evaluated high-resolution colourization on DIV2K, achieving **34.97 dB mean PSNR** across 10 images at native resolution without resizing or dataset-level training.

03. Multispectral Imaging for Environmental and Biological Sensing *2023 – 2025*

- Co-developed *custom multispectral imaging hardware and acquisition software*, integrating wavelength-controlled illumination, embedded control, image acquisition, calibration, and preprocessing.
- Developed *machine-learning pipelines for non-destructive soil characterization*, achieving R^2 **up to 0.989** for clay-silt-sand composition estimation and **92.86% mean accuracy** for USDA soil texture classification.
- Investigated *transmission-mode multispectral imaging and regression methods* for independent estimation of *Microcystis* cell concentration and ELISA-equivalent microcystin concentration.

Research Outputs: IEEE Access | ICATC | Soil Texture Code | Microcystis Code

Honors and Awards

Second Runner-Up, Manamperi Award – National award recognizing outstanding undergraduate engineering research projects in Sri Lanka; awarded for the final-year Multispectral Imaging project *2025*

Silver Award, Engineering Category – Sahasak Nimavum National Inventions and Innovations Competition; awarded for the Multispectral Imaging project *2024*

Top 10 Finalist, Predicta 1.0 – National AI and machine learning prediction challenge; selected among the top 10 from over 200 participants *2024*

Finalist, IEEE Electronic Design Competition – Advanced to the final round of the competition organized by the IEEE Sri Lanka Section *2023*

Second Place, ACES Hackathon – Healthcare category; posture-detection wearable garment presented at a university innovation hackathon *2023*

Fourth Place, IESL RoboGames – University category of the national robotics competition organized by the Institution of Engineers, Sri Lanka *2022*

Technical Skills

- **Programming:** Python, MATLAB, C/C++, Assembly
- **Machine Learning and Deep Learning:** PyTorch, TensorFlow, scikit-learn; regression, classification, generative modeling, model evaluation
- **Signal Processing:** Spectral and time-frequency analysis, wavelet analysis, empirical mode decomposition (EMD), MFCC, SciPy
- **Image Processing and Computational Imaging:** OpenCV, implicit neural representations (INRs), SIREN, image reconstruction, multispectral image processing
- **Embedded and Sensing Systems:** Raspberry Pi, ESP32, nRF5340, Zephyr RTOS, FreeRTOS, sensor integration, circuit design and calibration
- **Research and Development Tools:** Linux, Git, LaTeX, Altium Designer, EasyEDA

Academic Service and Outreach

AI Workshop Facilitation and Demonstrations – Supported AI sessions for pre-engineering students at the Diamond Jubilee AI Workshop and contributed to AI research demonstrations at the ESOFT Workshop *2025 – 2026*

Research Speaker and Demonstrator – Presented AI applications at Trinity College's AI for All event and demonstrated research at EngEx, Mayanz Expo, and DreamEx *2025*

Educational AI Outreach, MARC – Participated in AI educational outreach for school and university students in Hatton, Bandarawela, Kurunegala, and Vavuniya *2025 – 2026*

Student Chapter Service, IEEE RAS and IET on Campus – Supported robotics competitions, engineering workshops, and student engagement as a logistics team member and volunteer *2021 – 2023*

View Activities and Outreach

Academic References

1. **Prof. G. M. R. I. Godaliyadda**
   Ph.D., National University of Singapore; B.Sc. Eng. (Hons.), University of Peradeniya
   Professor, Department of Electrical and Electronic Engineering
   University of Peradeniya, Sri Lanka
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2. **Prof. H. M. V. R. Herath**
   Dr.-Ing., University of Paderborn, Germany; M.S., University of Miami, USA; B.Sc. Eng., University of Peradeniya
   Professor, Department of Electrical and Electronic Engineering
   University of Peradeniya, Sri Lanka
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3. **Prof. M. P. B. Ekanayake**
   Ph.D. in Applied Mathematics, Texas Tech University, USA; B.Sc. Eng. (Hons.), University of Peradeniya
   Professor, Department of Electrical and Electronic Engineering
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